
FACE
Prep

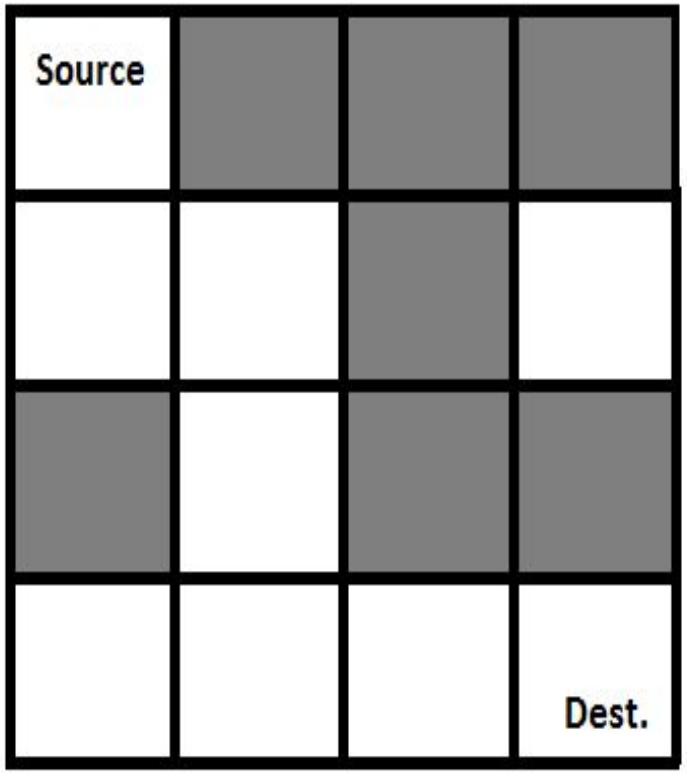
Maze Solving

- **Introduction**

  - Consider the maze to be a black and white image, with black pixels representing walls, and white pixels representing a path. Two white pixels are special, one being the entry to the maze and another exit.
  - Given such a maze, we want to find a path from entry to the exit.

- **Modelling the Maze**
- 0 -> Road
- 1 -> Wall
- 2 -> Maze entry
- 3 -> Maze exit
- 4 -> Cell part of the path from entry to exit

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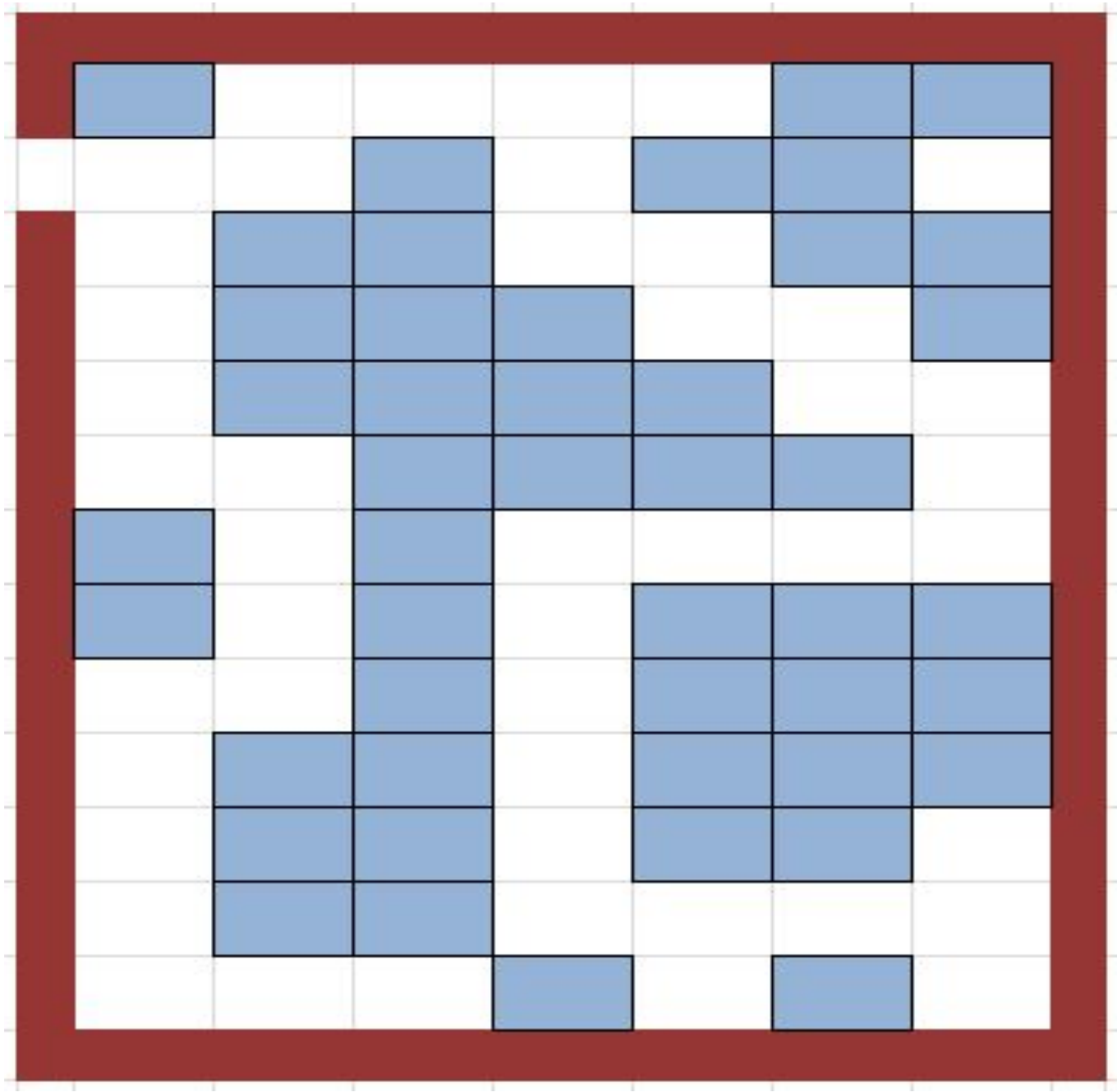


Gray blocks are dead ends (value = 0).

binary matrix representation of the above maze.

```
{ 1, 0, 0, 0}
{1, 1, 0, 1}
{0, 1, 0, 0}
{1, 1, 1, 1}
```

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```
1.  public class RatMaze {
2.     final int N = 4;
3.  void printSolution(int sol[][]){
4.        for (int i = 0; i < N; i++) {
5.           for (int j = 0; j < N; j++)
6.              System.out.print(" " + sol[i][j] + " ");
7.           System.out.println();
8.        }}
9.  boolean isSafe(int maze[][], int x, int y){
10.   return (x >= 0 && x < N && y >= 0 && y < N &&
      maze[x][y] == 1);
11.    }
12.  {
13.       int sol[][] = { { 0, 0, 0, 0 },
14.                       { 0, 0, 0, 0 },
15.                       { 0, 0, 0, 0 },
16.                       { 0, 0, 0, 0 } };
```

```
1.  if (solveMazeUtil(maze, 0, 0, sol) == false) {
2.           System.out.print("Solution doesn't exist");
3.           return false;
4.        }
5.        printSolution(sol);
6.        return true;
7.     }
8.  boolean solveMazeUtil(int maze[][], int x, int y,int sol[][]){
9.    if (x == N - 1 && y == N - 1) {
10.          sol[x][y] = 1;
11.          return true;
12.       }
13.   if (isSafe(maze, x, y) == true) {
14.  sol[x][y] = 1;
15.  if (solveMazeUtil(maze, x + 1, y, sol))
16.              return true;
17.  if (solveMazeUtil(maze, x, y + 1, sol))
18.              return true;
```

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```
  sol[x][y] = 0;
          return false;
      }
      return false;
   }

   public static void main(String args[]){
      RatMaze rat = new RatMaze();
      int maze[][] = { { 1, 0, 0, 0 },
                       { 1, 1, 0, 1 },
                       { 0, 1, 0, 0 },
                       { 1, 1, 1, 1 } };
      rat.solveMaze(maze);
   }
}
```

```
maze[][]={ {1, 1, 1, 1, 1},
           {0, 0, 1, 0, 1},
           {1, 1, 1, 1, 0},
           {1, 0, 0, 0, 1},
           {1, 1, 1, 1, 1}   };
```

Output:

```
1  0  0  0
1  1  0  0
0  1  0  0
0  1  1  1
```

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THANK YOU



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